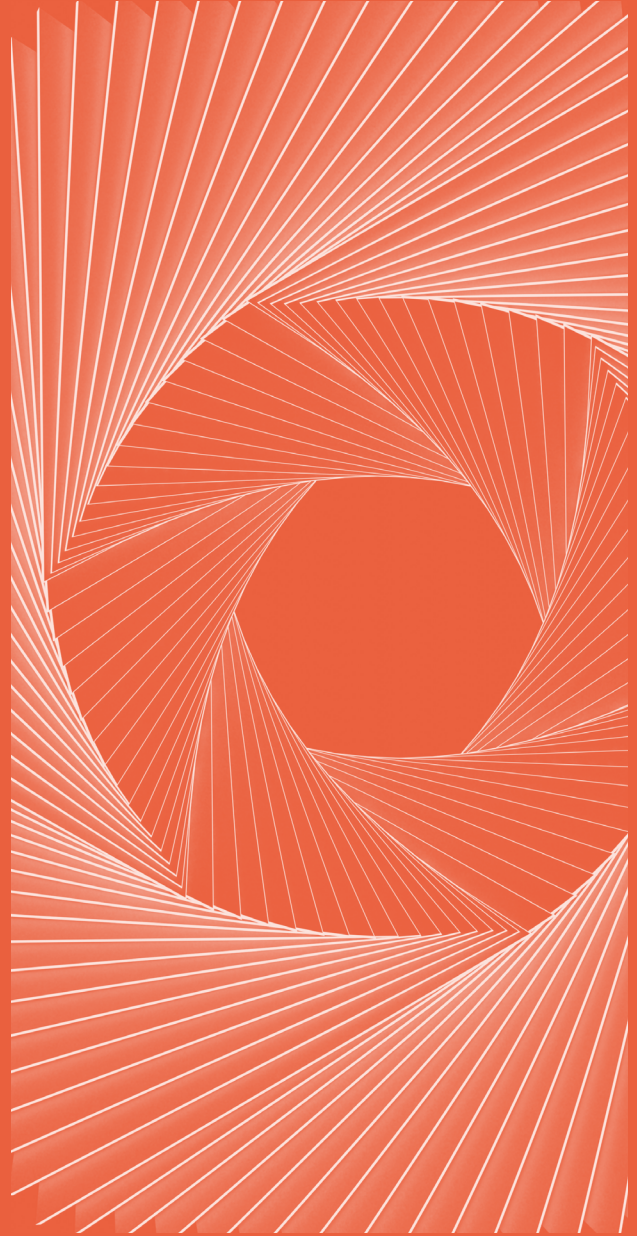


Market Data Recovery Feed Specification

Version 1.1



Updated September 2025



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3 Introduction

4 Supported Recovery Methods

9 IP Address and Port Reference

10 Revision History

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1.0 Introduction

This document describes how recipients of CIX market data feed may recover transmission of missed messages. This specification is intended as a companion to the CIX Market Data Feed Specification.

2.0 Supported Recovery Methods

CIX supports recovery of individual messages, as well as snapshot fast recovery of the current state of the books. Recipients may choose either recovery method at any time of the day depending on their technical requirements and preferences.

UDP Message Recovery

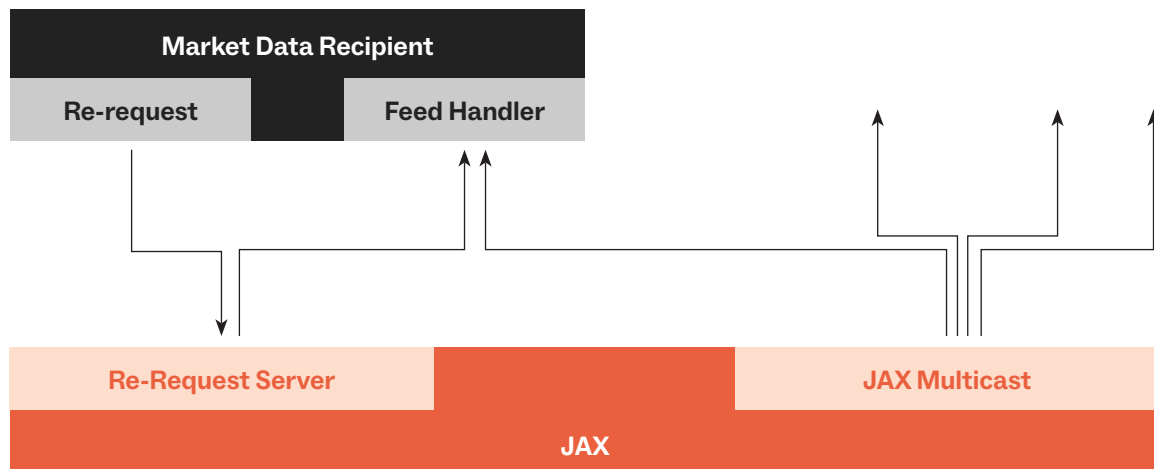
Missed messages may be re-requested using UDP. When a client sends a UDP rerequest, the rerequest server will reply with a single packet containing the requested messages. The packet will start at the requested Sequence, up to the requested Count of messages, capped by the number that would fit into a single packet. Note that the MTU of returned UDP packets will be capped at 1500 bytes.

Only a single UDP packet per request will be sent. Any messages that do not fit into that single UDP packet will not be sent until a subsequent request is made for them.

Response packets will be sent to clients at the IP address and Port contained in the source address IP header of the request packet. Request packets should be formed with the source port set to the desired return port. Users may, for example, choose their source port to be the same port as the live multicast feed destination port so that their feed handlers may process missed packets as they would normal feed packets.

There is no limit to the number of UDP re-requests. UDP recovery is intended to be suitable for full recovery even when a recipient starts their service late in the day.

Below is a diagram of the re-request server compared to the multicast feed.



UDP Rerequest Header

Note, the UDP re-request header is in the same format as the CIX multicast market data output feed header.

Field	Offset	Length	Type	Description
Market Day Identifier	0	9	Number	Must match current Market Day Identifier
Feed Identifier	9	1	Alpha	Must match Feed Identifier of rerequest server
Sequence	10	8	Integer	Sequence number for the first requested message
Count	18	2	Integer	Number of messages requested

Where to send re-request packets

Users will be provided with the IP addresses and ports of available UDP rerequest servers.

TCP Snapshots

TCP Snapshot recovery allows users to rapidly catch up to the CIX market data output feed intra-day.

Users will receive all **Market Event**, **Symbol Information**, and **Symbol State** messages.

However, for order specific messages, only messages related to orders currently on the books will be sent: **New Order Add**, **Order Partial Cancel**, **Order Cancel All**, and **Order Executed** messages.

Connecting to the TCP Snapshot Server

CIX has both a Primary and a Secondary TCP Snapshot server. Typically only one Snapshot service will be running at any given time. You will be provided with both Primary and Secondary IP addresses and ports. We recommend that client software first try the Primary IP address/port. If the Primary IP/port is unavailable, then the client software should next try the Secondary address.

Message Formats

With the exception of the End of Snapshot message, messages will be sent in the same format as the multicast market data output feed. Please see the Market Data Output feed specification for the details.

End of Snapshot Message

An **End of Snapshot** Message will be sent once the snapshot has completed.

Note that the **End of Snapshot** message will be sent inside the *TCP Transport Binary Data* ('S') message.

Field	Offset	Length	Type	Description
Message Type	0	1	Alpha	'X'
Sequence	1	8	Integer	Sequence number of the IQX multicast output feed corresponding to the last message sent, the previous message to this one, where this snapshot ends.

Handoff from Snapshot to CIX Multicast

Customers may need to subscribe to CIX Multicast prior to or simultaneously with connecting to the Snapshot service. Due to possible delays over TCP, by the time the **End of Snapshot** message is received, the IQX feed may be already sending subsequent messages. Customers will likely want to already have processed these messages, or have them in their UDP receive queue awaiting processing, or avail themselves of the UDP recovery for times when after switchover from snapshot there are still missing messages.

Connecting to CIX TCP Snapshot servers

Clients will be provided with Username, Passwords, and IP/port information for TCP Snapshot servers.

TCP Transport

TCP Snapshots are available over the CIX TCPOut protocol. TCPOut is a simple encapsulation layer for outpoud data dissemination.

Note that TCPOut is a separate protocol from the CIXmulticast/re-request protocol. TCPOut messages do not contain UDP headers.

Messages from Client to Server

Login Message

The client must login to the server before it will receive data.

Field	Offset	Length	Type	Description
Length	0	2	Integer	Length of the message (49)
Message Type	2	1	Alpha	'L'

Field	Offset	Length	Type	Description
User	3	6	Alpha	Username
Pass	9	10	Alpha	Password
Market Day Identifier	19	9	Integer	Unique identifier for trading day
Feed Identifier	28	1	Alpha	Unique code for feed
Sequence	29	20	Alpha	Requested starting sequence. This number is in ASCII. 0 indicates that the client wishes to start at the latest message.
				Note: Snapshot connections <i>must</i> specify 1 to login.

Heartbeat

Clients should send heartbeats to the server at least once per second. Connections will be terminated by the server if it stops receiving heartbeats.

Field	Offset	Length	Type	Description
Length	0	2	Integer	The length of this message (3)
Message Type	2	1	Alpha	'R'

From Server to Client

Successful Login

Field	Offset	Length	Type	Description
Length	0	2	Integer	Length of the message (33)
Message Type	2	1	Alpha	'A'
Market Day Identifier	3	9	Integer	Unique identifier for trading day
Feed Identifier	12	1	Alpha	Unique code for feed
Sequence	13	20	Alpha	ASCII sequence number of the next Binary Data ('S') message.

Login Reject

Field	Offset	Length	Type	Description
Length	0	2	Integer	Length of the message (4)
Message Type	2	1	Alpha	'J'
Reason Code	3	1	Alpha	Reject Reason: 'A' : Invalid User/Password 'B' : Invalid Sequence 'S' : Invalid Market Day/Feed Id

Heartbeat

The server will send heartbeats to the client whenever there is otherwise no data for a period of time. It is recommended that clients disconnect and attempt to reconnect in the event that they do not receive data from the server for one second.

Field	Offset	Length	Type	Description
Length	0	2	Integer	The length of this message (3)
Message Type	2	1	Alpha	'H'

Binary Data Message

Field	Offset	Length	Type	Description
Length	0	2	Integer	Length of the message
Message Type	2	1	Alpha	'S'
Data	3	-	Binary	Data message (Length - 3 bytes of data)

Example Binary Data Message

To illustrate how the Binary Data Message encapsulates IQX messages, here is an example of an encapsulated IQX *Symbol State Message*.

Note that the Binary Data Message 3 byte *header* is highlighted in **red**, and the encapsulated CIX message is in **blue**.

0×0000 | **18 00 53** 43 7C FF C7 9E AB 39 C7 5F A0 15 39 34

0×0010 | 33 32 41 46 00 00 00 00

- Length : 0×0018 (24)
 - 24 = Length of the Binary Data Message header (3 bytes) plus the length of the encapsulated Symbol State message (21 bytes).
- Message Type : 0×53 'S' (*Binary Data Message*)
- Data:
 - The encapsulated Symbol State message (CIX message type 'C')

Type	Symbol ID	Timestamp	Symbol Code	State	Short Sale Restriction	Info
C	31999	1558350780340870855	9432	A	F	

3.0 IP Address and Port Reference

The following table lists the current IP and port assignments for available recovery services.

Each consumer get their own data:

- For UDP, a single UDP packet response is sent back to the UDP requestor's IP/port.
- For TCP, each client will get Username/Password credentials. They then connect to the TCP port and receive TCP Snapshot responses. These TCP snapshot sessions are short lived, usually taking only a few seconds. Each instance of the Re-requestor application can handle multiple concurrent TCP sessions.

UDP Rerequest

Site	Feed Identifier	IP	Port
PROD	A - ASPEN	71.15.52.36	49011
PROD	V - ASPEN VERT	71.15.52.36	49012
PROD	M - MIDPOINT	71.15.52.36	49013
UAT	B - ASPEN	71.15.52.38	49111
UAT	W - ASPEN VERT	71.15.52.38	49112
UAT	N - MIDPOINT	71.15.52.38	49113

TCP Snapshot

Site	Feed Identifier	IP	Port
PROD	A - ASPEN	71.15.52.36	49001
PROD	V - ASPEN VERT	71.15.52.36	49002
PROD	M - MIDPOINT	71.15.52.36	49003
UAT	B - ASPEN	71.15.52.38	49101
UAT	W - ASPEN VERT	71.15.52.38	49102
UAT	N - MIDPOINT	71.15.52.38	49103

4.0 Revision History

Version	Date	Change
1.0	August 27, 2025	Initial Revision
1.1	September 15, 2025	Updated for IP Address and Port numbers



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